

Class-Imbalance Benchmarks for Whole-Slide Images and Patches

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Abstract

Class imbalance in computational pathology appears both as a native property of clinical cohorts and as an experimental factor that can be controlled by resampling. This report studies both forms in patch-level and whole-slide-image (WSI) feature-bag benchmarks. TCGA-UT provides a pan-cancer setting in which training imbalance is varied artificially by power-law sampling while validation and test splits remain case-disjoint and fixed. BRACS provides an external breast-lesion setting in which the native distribution over seven histologic subtypes is measured first and used only if it is meaningfully imbalanced; otherwise the same power-law severity sweep is applied. Across both datasets, frozen Virchow2 features and a fixed evidence budget separate imbalance mechanisms from encoder training. The current TCGA-UT results show that method rankings are largely preserved across constructed severities while balanced accuracy declines monotonically as rare-class support diminishes; macro F1 is more stable between $\lambda = 0.5$ and $\lambda = 1.0$ before dropping at $\lambda = 1.5$. BRACS’s native seven-subtype distribution measures approximately 2.25:1 after preparation—below the minimum ratio required for a meaningful native imbalance benchmark—so the same power-law severity sweep is applied instead. On BRACS, OKO retains its patch-level lead across all three constructed severities and the absolute severity decline is smaller than on TCGA-UT, while the WSI-bag regime shows non-monotone responses and substantially higher between-seed variance, consistent with the constraints of the smaller 108-slide training pool and the greater difficulty of fine-grained seven-class breast-lesion subtyping.

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1 Introduction

A prior class-imbalance study on TCGA-UT compared mitigation methods under the dataset’s native, moderately imbalanced slide distribution (head-to-tail $\approx 28:1$) across 32 cancer types (Queisler, 2026). Native distributions are realistic but confound several factors simultaneously: how often each class appears, tumor morphology, tissue diversity, slide acquisition, and the representation geometry of the frozen encoder. A controlled construction isolates one factor—training-support frequency—by imposing known power-law distributions while holding validation and test data fixed. A native external dataset, in contrast, tests whether the same benchmark machinery remains meaningful when the class distribution is not designed by the experimenter.

TCGA-UT is a suitable test bed for this isolation because it provides slide labels and patch-level evidence from diagnostic whole-slide images in The Cancer Genome Atlas (Tomczak et al., 2015; Komura et al., 2022). Controlling only the imbalance severity under a fixed native-prevalence class ordering allows the question to be posed cleanly: do mitigation methods maintain their relative advantage as the imbalance between frequent and rare cancer types intensifies along the same ranking?

BRACS adds a complementary native benchmark. It is an H&E breast-pathology dataset with 547 WSIs and 4,539 regions of interest (ROIs) from 189 patients, annotated by consensus of three board-certified pathologists into three lesion types and seven subtypes (Brancati et al., 2022). The atypical categories are clinically important and comparatively scarce, making BRACS a natural class-imbalance setting without imposing power-law resampling. Unlike TCGA-UT, BRACS is organ-specific, has fewer labels, and provides lesion-focused ROI annotations; these differences make it useful as an external check rather than a duplicate of the pan-cancer experiment.

The report therefore contains two benchmark roles. TCGA-UT evaluates the same method families as Queisler (2026) under three constructed training distributions that vary only $\lambda \in \{0.5, 1.0, 1.5\}$, the imbalance-severity parameter. BRACS evaluates the same patch and WSI-bag method families on its native seven-subtype distribution. Patch-level methods include cross-entropy, class weighting, balanced sampling, focal loss, Scholz-style metric losses, CFAL, OKO set learning, and ProGAN synthetic augmentation (Scholz et al., 2024). WSI-bag methods include cross-entropy, class weighting, balanced sampling, focal loss, RankMix, SC-MIL, and MDE-MIL (Chen and Lu, 2023; Juyal et al., 2024; Ling et al., 2025). Patch and WSI-bag methods are reported separately because they consume different prediction units and answer regime-specific questions.

2 Dataset Construction

The two datasets serve different roles. TCGA-UT is used for a controlled severity sweep in which the training distribution is designed by the experimenter. BRACS is first treated as a native external benchmark: its observed seven-subtype distribution is measured and used directly if it provides a meaningful class-imbalance setting. If the prepared BRACS distribution is too close to balanced to test imbalance mitigation, the same power-law construction used for TCGA-UT is applied to BRACS instead, with $\lambda \in \{0.5, 1.0, 1.5\}$.

2.1 TCGA-UT Controlled Severity Benchmark

The construction shares the TCGA-UT case-disjoint split protocol with Queisler (2026). For each of three random seeds, the validation and test splits are held fixed, and artificial imbalance is applied only to the training split. The assignment is patient-disjoint: all slides and feature rows associated with one TCGA participant remain in exactly one partition. This isolates

the effect of training-support frequency: model selection and evaluation are comparable across imbalance parameters and mitigation methods without patient leakage. Strictly, the controlled lever is the per-class training-support *distribution*: because each λ selects a different number of slides per class from the shared pool, the specific slides realized for a class also change across λ . Differences observed across severities therefore reflect both the change in class counts and, secondarily, which slides happen to be sampled; the fixed validation and test splits and the shared slide pool limit but do not fully remove this slide-composition confound.

All 32 cancer types are retained. Let $C = 32$, let $\mathcal{D}_{\text{train}}$ denote the seed-specific training slide pool, and let a_c be the number of available training slides for class c . The constructed training split must satisfy

$$1 \leq n_c \leq a_c, \quad \sum_{c=1}^C n_c = N, \quad c \in \{1, \dots, C\}. \quad (1)$$

Every class receives at least one slide, no class is oversampled beyond its available pool, and the total training set is fixed at N slides shared across all imbalance parameters and seeds.

After slide selection, the input evidence is capped. Patch classifiers use up to 30 deterministic patches per slide. WSI-bag classifiers use up to 30 deterministic frozen feature instances per slide. In both regimes, Virchow2 features remain frozen (Vorontsov et al., 2024; Zimmermann et al., 2024), represented as the concatenation of the CLS token and the mean of the patch tokens (2,560-dimensional vectors). The cap aligns the input budget across regimes and prevents slides with more extracted tumor patches from dominating the training objective.

2.2 Power-Law Training Support

For an ordered class list $(c_1, \dots, c_C)$, the intended share for rank i under imbalance parameter λ is

$$p_i(\lambda) = \frac{i^{-\lambda}}{\sum_{j=1}^C j^{-\lambda}}. \quad (2)$$

When $\lambda = 0$ the intended distribution is uniform over ranks; as λ increases, early-ranked classes receive more training support and late-ranked classes receive less. All constructed splits use the *native-prevalence order*: classes are ranked by their TCGA-UT training-split slide count, so rank 1 is the most frequent cancer type and rank $C = 32$ the least frequent. The power-law form is imposed to obtain controlled, monotonic severity steps; it is not a fit to the native TCGA-UT distribution, which is only moderately imbalanced ($\approx 28:1$) and not well described by a single power law.

Raw targets $N p_i(\lambda)$ are converted to integer counts by flooring and assigning residual slides by largest fractional remainder. Each class first receives one guaranteed slide; the remaining $N - C$ slides are then allocated according to Equation (2). A regime is accepted only when every rounded target satisfies Equation (1).

2.3 Feasible Pool Size and Realized Profiles

The pool size N must be small enough that no class target exceeds its available support across all evaluated imbalance parameters. Define the maximum feasible pool for a given λ and seed as

$$T^*(\lambda) = \min_c \frac{a_c}{p_c(\lambda)}. \quad (3)$$

Equation (3) is a continuous bound on the real-valued targets $N p_c(\lambda)$ and does not by itself account for the guaranteed-one floor, flooring, and largest-remainder steps of the integer allocation. The operational pool size is therefore the largest total whose *integer* allocation satisfies Equation (1) on every seed, found by a feasibility search that runs the full rounding rule and

rejects any total that overruns a class capacity. The continuous bound $\lfloor \min_s T_s^*(0.5) \rfloor$ and this integer-verified value coincide for the TCGA-UT splits, so reporting the closed-form bound is exact here; the integer check is what guarantees feasibility in general. The binding regime is $\lambda = 0.5$ rather than the steepest: the flat power-law profile at $\lambda = 0.5$ requires every class—including rare ones with limited available slides—to contribute a non-trivial share, saturating the rarest class’s supply at a lower total N than the peaked profiles at $\lambda \geq 1.0$. For the TCGA-UT training splits, this yields $N = 1,132$ slides, approximately 18% of the full available training pool.

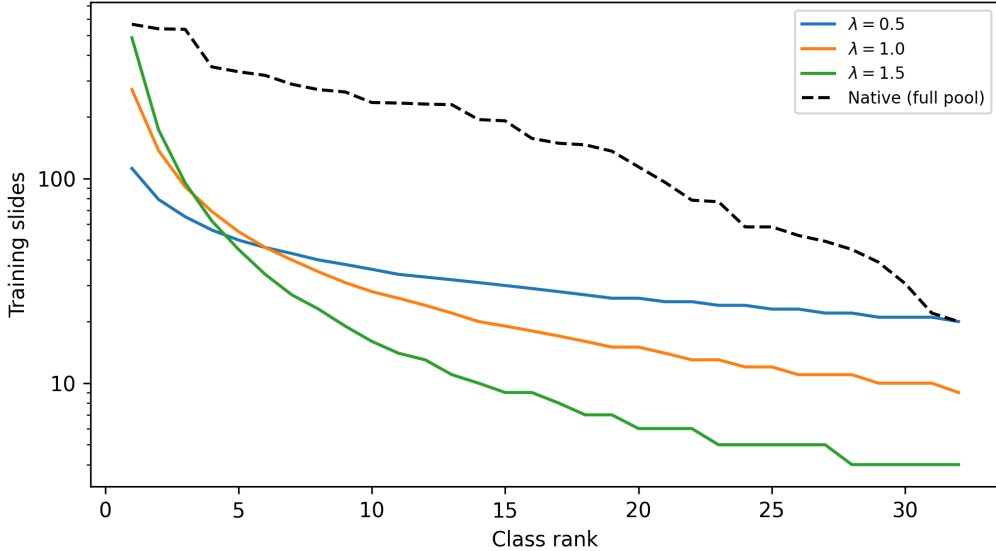


Figure 1: Constructed training support by class rank under the three evaluated imbalance parameters, with the native full-pool distribution shown as a reference. Classes are ordered by native prevalence (rank 1 = most frequent). The vertical axis (training slides per class) is log-scaled.

The three λ values produce head-to-tail ratios of approximately 6:1 ($\lambda = 0.5$), 30:1 ($\lambda = 1.0$), and 122:1 ($\lambda = 1.5$) (Figure 1). The $\lambda = 0.5$ setting places the constructed distribution well below the native TCGA-UT ratio of approximately 28:1; $\lambda = 1.0$ reproduces near-native severity; and $\lambda = 1.5$ is substantially more skewed than anything observed in the native data.

2.4 BRACS Native Benchmark

BRACS is prepared as a second benchmark because it changes the data regime without changing the evaluation question. It is an organ-specific H&E breast-lesion dataset rather than a pan-cancer dataset, and its annotations are lesion-subtype labels rather than cancer-type labels. The dataset contains 547 WSIs and 4,539 ROIs from 189 patients, annotated by consensus of three board-certified pathologists (Brancati et al., 2022). The labels form three lesion types—benign, atypical, and malignant—and seven subtypes: normal tissue (N), pathological benign (PB), usual ductal hyperplasia (UDH), flat epithelial atypia (FEA), atypical ductal hyperplasia (ADH), ductal carcinoma in situ (DCIS), and invasive carcinoma (IC).

The benchmark uses the seven-subtype target because it is the level at which class imbalance is most relevant. The atypical categories FEA and ADH are clinically important precancerous lesions and are less common than many benign or malignant samples, so collapsing BRACS to the three lesion types would hide part of the rare-class problem. Patient identifiers from the BRACS summary sheet define the split unit. The prepared benchmark therefore enforces

patient-disjoint train, validation, and test partitions across three seeds, matching the leakage-control standard used for TCGA-UT.

BRACS evidence is derived from the ROI set. Each annotated ROI is tiled deterministically into 256×256 RGB patches, capped at 30 tiles per ROI to match the evidence budget used for TCGA-UT. Patch methods consume one frozen Virchow2 feature per ROI tile. The WSI-bag pipeline consumes grouped ROI-tile features as bounded bags, preserving the same patch-versus-bag comparison while avoiding an additional tissue-detection step on full WSIs. This choice prioritizes label fidelity: all training instances originate from pathologist-selected lesion regions rather than from unlabeled background tissue.

After patch extraction, 387 of the 547 published WSIs contribute at least one annotated ROI tile; the remaining slides have no tiled ROI annotations available. The prepared benchmark covers 151 patients, the full 4,539 ROI metadata records, and 98,282 extracted patches (Table 1). The seven-subtype WSI support spans PB (196 slides) at the head to DCIS (87 slides) at the tail, a ratio of approximately 2.25:1. This falls below the minimum ratio of 5:1 required for a meaningful native imbalance benchmark, so the benchmark switches to the same three power-law severity levels as TCGA-UT. The feasible pool size is 108 WSIs, yielding constructed head-to-tail ratios of approximately 3:1 ($\lambda = 0.5$), 6:1 ($\lambda = 1.0$), and 14:1 ($\lambda = 1.5$)—a narrower and milder range than TCGA-UT, reflecting the smaller label set and shallower per-class WSI supply.

Property	Value
Patients	151
WSIs with ROI tiles	387
ROI metadata rows	4,539
ROI tiles	98,282
Subtype labels	N, PB, UDH, FEA, ADH, DCIS, IC
WSI support range	87–196
WSI imbalance ratio	$\approx 2.3:1$

Table 1: BRACS dataset properties used by the native benchmark. Generated rows add the prepared ROI-tile counts and measured subtype imbalance after staging.

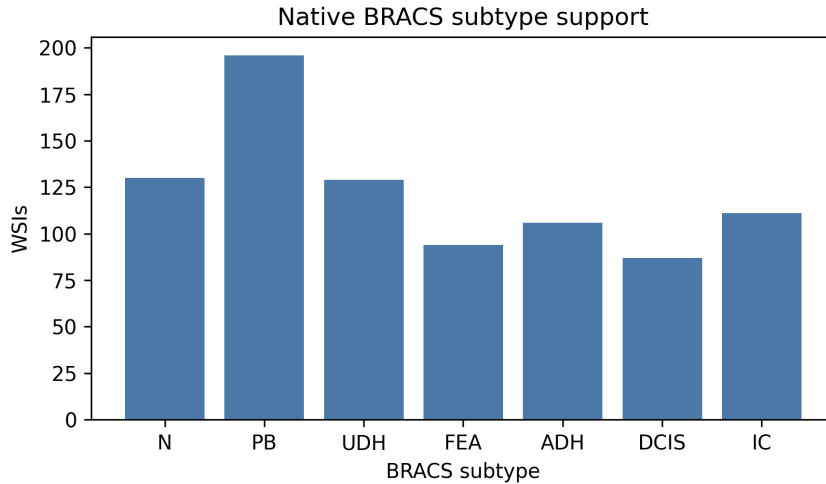


Figure 2: Native BRACS WSI support by seven-subtype label after patient-disjoint preparation. If the observed support range is not sufficiently imbalanced, BRACS is evaluated with the same three power-law severity settings as TCGA-UT rather than as a native imbalance benchmark.

3 Methods

The method taxonomy follows the companion benchmark, but this report uses it to study controlled training distributions rather than only the native TCGA-UT distribution. Cross-entropy is the no-mitigation reference. Data-level methods change minibatch exposure through balanced sampling or synthetic augmentation. Algorithm-level methods change the per-example loss through class weighting, focal loss, metric-derived objectives, prototype-affinity objectives, or OKO set supervision. WSI-specific methods change how slide bags are represented or mixed.

Family	Methods	Regime
No-mitigation baseline	CE / MIL CE	Patch, WSI bag
Data-level sampling and augmentation	Balanced sampling, RankMix	Patch, WSI bag
Algorithm-level losses	Weighted CE, Focal loss, CFAL	Patch (CFAL); Patch and WSI bag (others)
Hybrid sampling-loss	CE + soft F1, CE + soft MCC, OKO, MDE-MIL	Patch (first three, OKO); WSI bag (MDE-MIL)
Synthetic data generation	ProGAN augmentation	Patch
Representation and architecture	SC-MIL	WSI bag

Table 2: Method families evaluated under the TCGA-UT controlled severity benchmark and the BRACS external benchmark, following the taxonomy of Queisler (2026). Patch and WSI-bag methods are compared only within their own input regime.

Full method definitions follow Queisler (2026) and Scholz et al. (2024). RankMix, SC-MIL, and MDE-MIL are WSI-bag methods whose mechanisms operate on bags rather than isolated patch embeddings (Chen and Lu, 2023; Juyal et al., 2024; Ling et al., 2025).

4 Evaluation Protocol

Each TCGA-UT constructed split is indexed by seed and imbalance parameter $\lambda \in \{0.5, 1.0, 1.5\}$. Hyperparameters are selected independently for every method and every λ : for each candidate configuration the validation macro F1 is averaged over the three seeds, and the configuration with the highest mean is chosen, with ties broken by mean validation balanced accuracy. A single configuration is therefore fixed per method and λ (shared across seeds) and is not reused across λ . Because tuning is performed separately at each severity, the reported figures reflect each method at its per- λ best, which can partly absorb the effect of imbalance severity; the comparison thus targets achievable rather than fixed-configuration performance. Selected configurations are evaluated on the fixed test split, and results are reported as mean $\pm$ standard deviation across seeds. Macro F1 is the primary discrimination metric because it weights classes equally. Balanced accuracy and accuracy are reported as complementary discrimination metrics.

BRACS follows the same validation-selection rule, but the regime index depends on the prepared distribution. If the native seven-subtype support is meaningfully imbalanced, one native configuration is selected per method by averaging validation macro F1 across the three patient-disjoint seeds. If the native support is not a useful class-imbalance setting, BRACS is switched to the same $\lambda \in \{0.5, 1.0, 1.5\}$ power-law construction as TCGA-UT, and tuning is performed per λ .

The comparison is intentionally regime-specific. Patch-feature methods are compared only with other patch-feature methods, and WSI-bag methods are compared only with other WSI-bag methods. The two regimes share the case-disjoint splits, frozen Virchow2 representation family, and 30-instance evidence budget, but they differ in prediction unit and aggregation mechanism.

Calibration is evaluated alongside discrimination using negative log likelihood (NLL), multiclass Brier score (Brier, 1950), and expected calibration error (ECE; Naeini et al., 2015; Guo et al., 2017). For each method and seed, a single temperature is fit by minimizing NLL on the validation split and applied unchanged to the test score matrix before softmax (Guo et al., 2017). Tables 9 and 10 report ECE before and after temperature scaling; full NLL and Brier metrics are given in Tables 13 and 14.

5 Results

The analysis asks how methods respond as imbalance severity increases from $\lambda = 0.5$ to $\lambda = 1.5$ under the native-prevalence class order. The realized support profiles are summarised in Table 11 (Appendix A); Tables 3 and 4 compare mitigation methods within the patch-feature and WSI-bag regimes.

Method	$\lambda = 0.5$		$\lambda = 1.0$		$\lambda = 1.5$		Native	
	BAcc	F1	BAcc	F1	BAcc	F1	BAcc	F1
Weighted CE	0.721 ± 0.012	0.693 ± 0.006	0.704 ± 0.012	0.693 ± 0.016	0.634 ± 0.006	0.641 ± 0.009	0.767 ± 0.006	0.774 ± 0.005
Focal loss	0.718 ± 0.011	0.692 ± 0.007	0.698 ± 0.016	0.689 ± 0.018	0.630 ± 0.005	0.641 ± 0.009	0.758 ± 0.007	0.765 ± 0.009
CFAL	0.724 ± 0.009	0.697 ± 0.007	0.710 ± 0.014	0.704 ± 0.015	0.632 ± 0.013	0.645 ± 0.014	0.789 ± 0.007	0.798 ± 0.009
CE + soft F1	0.721 ± 0.012	0.685 ± 0.004	0.713 ± 0.007	0.686 ± 0.014	0.638 ± 0.021	0.630 ± 0.019	0.781 ± 0.002	0.780 ± 0.003
CE + soft MCC	0.719 ± 0.006	0.690 ± 0.009	0.698 ± 0.010	0.684 ± 0.009	0.634 ± 0.019	0.633 ± 0.018	0.772 ± 0.004	0.770 ± 0.005
Balanced sampling	0.719 ± 0.012	0.691 ± 0.007	0.701 ± 0.015	0.688 ± 0.018	0.631 ± 0.012	0.635 ± 0.013	0.769 ± 0.007	0.771 ± 0.014
OKO	0.722 ± 0.012	0.689 ± 0.008	0.712 ± 0.011	0.694 ± 0.018	0.644 ± 0.015	0.648 ± 0.014	0.791 ± 0.005	0.788 ± 0.003
ProGAN augmentation	0.729 ± 0.003	0.698 ± 0.001	0.712 ± 0.003	0.700 ± 0.015	0.624 ± 0.016	0.636 ± 0.016	0.760 ± 0.005	0.766 ± 0.002

Table 3: Patch-feature test results (balanced accuracy and macro F1) for validation-selected configurations under constructed training distributions. The Native column reports the companion full-pool benchmark (Queisler, 2026) (native $\approx 28:1$ imbalance, complete training pool) as a reference; it is not part of the constructed experiment.

Patch-feature balanced accuracy shows a consistent, monotone decline as imbalance intensifies, whereas macro F1 is comparatively flat between $\lambda = 0.5$ and $\lambda = 1.0$ —rising slightly for several methods (CFAL, CE+soft-F1, OKO, ProGAN)—before dropping at $\lambda = 1.5$ (Table 3). ProGAN augmentation numerically leads in balanced accuracy at the mildest severity but reverses to last at the harshest, undergoing the steepest decline of any patch method; OKO and CE+soft-F1 are the most stable and numerically retain the lead at $\lambda = 1.5$. With a within-severity spread of only one to two percentage points and overlapping standard deviations, these orderings should be read as numerical rather than statistically separated. Focal loss is typically among the weakest of the remaining methods, though by margins within seed-to-seed variation. The key observation is one of scale: the spread across methods at any fixed λ is only one to two percentage points, whereas the severity-induced decline from $\lambda = 0.5$ to $\lambda = 1.5$ is roughly eight to eleven percentage points—choosing the right severity matters far more than choosing the best method. The native full-pool benchmark (rightmost block) sits above even the mildest constructed severity, despite carrying a larger imbalance ($\approx 28:1$) than $\lambda = 0.5$ ($\approx 6:1$); because it draws on the complete training pool rather than the constructed subset, this gap suggests training-set size—rather than imbalance—is the dominant lever on absolute patch performance.

The WSI-bag regime shows higher absolute performance than the patch regime (Table 4), roughly ten balanced-accuracy points above patch methods at equivalent imbalance. MDE-MIL numerically leads at mild and moderate severities, but RankMix is clearly ahead at the harshest regime ($\lambda = 1.5$, BAcc 0.765 vs 0.730) and is the most severity-robust method overall, while focal loss is the least stable. MDE-MIL’s advantage over simple loss weighting erodes at $\lambda = 1.5$, where it falls narrowly below Weighted CE in balanced accuracy and ties it in macro F1—consistency regularisation provides diminishing returns as rare-class support becomes critically scarce. RankMix is consistently among the strongest WSI-bag methods and is the most robust at

Method	$\lambda = 0.5$		$\lambda = 1.0$		$\lambda = 1.5$		Native	
	BAcc	F1	BAcc	F1	BAcc	F1	BAcc	F1
Weighted CE	0.823 $\pm$ 0.019	0.793 $\pm$ 0.017	0.799 $\pm$ 0.008	0.787 $\pm$ 0.007	0.736 $\pm$ 0.027	0.731 $\pm$ 0.021	0.833 $\pm$ 0.019	0.835 $\pm$ 0.011
Focal loss	0.821 $\pm$ 0.013	0.792 $\pm$ 0.012	0.801 $\pm$ 0.009	0.792 $\pm$ 0.007	0.703 $\pm$ 0.029	0.708 $\pm$ 0.031	0.823 $\pm$ 0.019	0.824 $\pm$ 0.008
Balanced sampling	0.809 $\pm$ 0.014	0.776 $\pm$ 0.010	0.797 $\pm$ 0.010	0.785 $\pm$ 0.018	0.724 $\pm$ 0.042	0.720 $\pm$ 0.040	0.829 $\pm$ 0.010	0.831 $\pm$ 0.009
RankMix	0.830 $\pm$ 0.016	0.793 $\pm$ 0.015	0.817 $\pm$ 0.024	0.795 $\pm$ 0.020	0.765 $\pm$ 0.029	0.751 $\pm$ 0.019	0.864 $\pm$ 0.002	0.854 $\pm$ 0.008
SC-MIL	0.813 $\pm$ 0.015	0.781 $\pm$ 0.014	0.800 $\pm$ 0.013	0.787 $\pm$ 0.008	0.745 $\pm$ 0.022	0.743 $\pm$ 0.015	0.837 $\pm$ 0.013	0.839 $\pm$ 0.008
MDE-MIL	0.836 $\pm$ 0.013	0.805 $\pm$ 0.012	0.825 $\pm$ 0.009	0.814 $\pm$ 0.007	0.730 $\pm$ 0.009	0.731 $\pm$ 0.014	0.849 $\pm$ 0.014	0.865 $\pm$ 0.011

Table 4: WSI-bag test results (balanced accuracy and macro F1) for validation-selected configurations under constructed training distributions. The Native column is the companion full-pool benchmark (Queisler, 2026) reference, as in Table 3.

$\lambda = 1.5$; SC-MIL remains competitive but does not uniformly outperform loss-based baselines—at $\lambda = 0.5$ it sits below both Weighted CE and focal loss. Because many gaps are one to two percentage points with overlapping standard deviations, these orderings are best read as numerical leads rather than statistically separated outcomes. As in the patch regime, the native full-pool benchmark (rightmost block) exceeds even $\lambda = 0.5$ despite its larger imbalance, again pointing to training-set size rather than imbalance severity as the main driver of absolute performance.

5.1 BRACS Power-Law Breast-Lesion Benchmark

The BRACS benchmark is reported separately because its target is seven-class breast-lesion subtyping rather than 32-class pan-cancer recognition. The native distribution proved too balanced (2.25:1) for a native imbalance benchmark, so Tables 5 and 6 report the power-law severity sweep applied instead. Scores are not directly comparable to TCGA-UT: the label set, task difficulty, and pool size all differ.

Method	$\lambda = 0.5$		$\lambda = 1.0$		$\lambda = 1.5$	
	BAcc	F1	BAcc	F1	BAcc	F1
Weighted CE	0.395 $\pm$ 0.030	0.399 $\pm$ 0.036	0.387 $\pm$ 0.031	0.386 $\pm$ 0.034	0.352 $\pm$ 0.021	0.341 $\pm$ 0.021
Focal loss	0.397 $\pm$ 0.035	0.399 $\pm$ 0.040	0.387 $\pm$ 0.033	0.385 $\pm$ 0.036	0.356 $\pm$ 0.019	0.346 $\pm$ 0.018
CFAL	0.392 $\pm$ 0.053	0.397 $\pm$ 0.057	0.394 $\pm$ 0.030	0.392 $\pm$ 0.033	0.360 $\pm$ 0.026	0.346 $\pm$ 0.029
CE + soft F1	0.392 $\pm$ 0.043	0.397 $\pm$ 0.048	0.396 $\pm$ 0.033	0.395 $\pm$ 0.034	0.354 $\pm$ 0.007	0.349 $\pm$ 0.009
CE + soft MCC	0.401 $\pm$ 0.046	0.401 $\pm$ 0.059	0.378 $\pm$ 0.038	0.376 $\pm$ 0.031	0.349 $\pm$ 0.021	0.340 $\pm$ 0.019
Balanced sampling	0.395 $\pm$ 0.041	0.398 $\pm$ 0.048	0.380 $\pm$ 0.015	0.378 $\pm$ 0.019	0.363 $\pm$ 0.016	0.355 $\pm$ 0.018
OKO	0.423 $\pm$ 0.048	0.421 $\pm$ 0.040	0.418 $\pm$ 0.041	0.410 $\pm$ 0.045	0.400 $\pm$ 0.008	0.386 $\pm$ 0.019
ProGAN augmentation	0.393 $\pm$ 0.032	0.396 $\pm$ 0.041	0.385 $\pm$ 0.022	0.384 $\pm$ 0.025	0.356 $\pm$ 0.026	0.345 $\pm$ 0.025

Table 5: BRACS patch-feature test results (balanced accuracy and macro F1) under the three power-law severity levels, mean $\pm$ standard deviation across patient-disjoint seeds. Pool size: 108 WSIs; head-to-tail ratios approximately 3:1, 6:1, and 14:1 for $\lambda = 0.5$, 1.0, and 1.5.

In the patch regime, all methods cluster within approximately one percentage point of each other at the mildest severity (0.392–0.401 balanced accuracy), with OKO as the only clear outlier at 0.423 (Table 5). The severity decline is 2.3 to 4.3 percentage points across methods—roughly half the decline observed on TCGA-UT—consistent with the narrower achievable imbalance range (3:1 to 14:1) and the smaller pool. OKO is the most stable method (Δ BAcc = 2.3 pp) and the only one that separates from the cluster at all three severities; CE+soft-F1 and CFAL are the next most stable. ProGAN augmentation does not reverse to worst at the harshest severity as it did on TCGA-UT; its decline (3.7 pp) matches that of the loss-based baselines, likely because tail classes retain enough per-class patches to support synthetic augmentation across all three severity levels.

Method	$\lambda = 0.5$		$\lambda = 1.0$		$\lambda = 1.5$	
	BAcc	F1	BAcc	F1	BAcc	F1
Weighted CE	0.479 ± 0.090	0.473 ± 0.090	0.540 ± 0.022	0.523 ± 0.021	0.432 ± 0.054	0.416 ± 0.058
Focal loss	0.509 ± 0.012	0.509 ± 0.010	0.516 ± 0.008	0.520 ± 0.019	0.451 ± 0.038	0.419 ± 0.050
Balanced sampling	0.508 ± 0.031	0.504 ± 0.026	0.470 ± 0.050	0.462 ± 0.066	0.437 ± 0.061	0.431 ± 0.052
RankMix	0.478 ± 0.100	0.480 ± 0.086	0.499 ± 0.067	0.494 ± 0.066	0.448 ± 0.033	0.425 ± 0.060
SC-MIL	0.497 ± 0.045	0.511 ± 0.045	0.532 ± 0.041	0.525 ± 0.050	0.459 ± 0.072	0.447 ± 0.047
MDE-MIL	0.472 ± 0.032	0.477 ± 0.019	0.517 ± 0.012	0.522 ± 0.014	0.455 ± 0.091	0.432 ± 0.092

Table 6: BRACS WSI-bag test results under the three power-law severity levels, mean $\pm$ standard deviation across patient-disjoint seeds. Pool size and severity levels as in Table 5.

The WSI-bag regime differs from TCGA-UT in two important respects. First, severity responses are non-monotone: Weighted CE and SC-MIL both peak at $\lambda = 1.0$ (0.540 and 0.532, respectively) before falling at $\lambda = 1.5$, and the same pattern recurs for MDE-MIL and focal loss. Second, between-seed standard deviations reach ± 0.09 – 0.10 , substantially higher than the ± 0.02 – 0.05 typical on TCGA-UT. RankMix, the most severity-robust method on TCGA-UT, shows the highest variance at mild severity (0.478 ± 0.100) and does not lead at any BRACS severity level. SC-MIL is numerically ahead at $\lambda = 1.0$ (BAcc 0.532) and $\lambda = 1.5$ (BAcc 0.459), though the overlapping standard deviations mean these differences are not statistically separated.

Tail-Class Behaviour. Aggregate macro F1 and balanced accuracy can mask whether a method helps rare cancer types or merely preserves frequent ones. To separate these, classes are grouped into support tiers by native prevalence—head (the 8 most frequent), body (the middle 16), and tail (the 8 rarest)—and per-class test recall is plotted against the realized training support of each class (Figure 3). Tables 7 and 8 report tier-resolved macro F1 per severity.

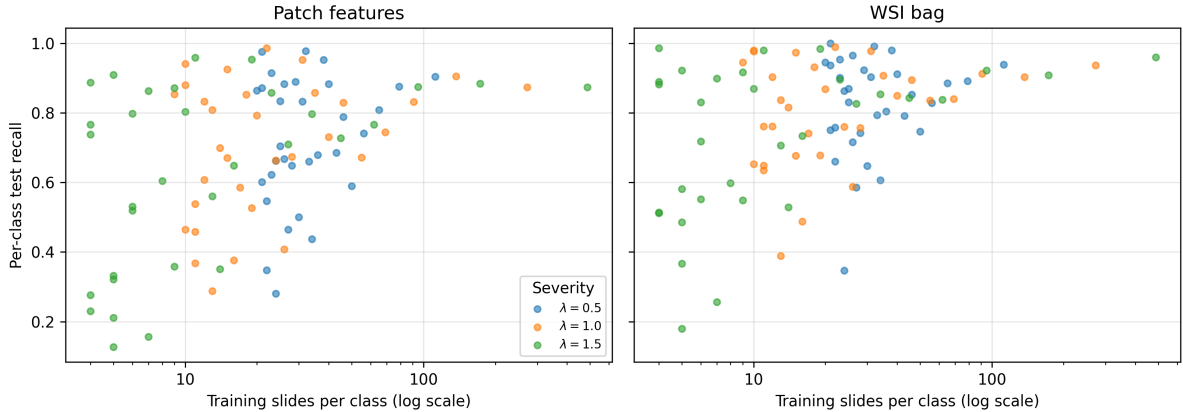


Figure 3: Per-class test recall against training slides per class (log scale), one panel per regime. Each point is a cancer type at one severity, averaged over methods and seeds; colour encodes the imbalance parameter λ . The horizontal axis is log-scaled.

The support–recall relationship makes the locus of the severity effect explicit (Figure 3): per-class recall is near-saturated (≈ 0.9) for high-support classes and degrades sharply for low-support classes, with the $\lambda = 1.5$ points dominating the low-support, low-recall corner. The tier-resolved tables confirm that the aggregate decline is concentrated in the rare classes. Head-tier macro F1 is essentially flat across severity—patch head F1 holds near 0.79 and only

Method	$\lambda = 0.5$			$\lambda = 1.0$			$\lambda = 1.5$		
	Head	Body	Tail	Head	Body	Tail	Head	Body	Tail
Weighted CE	0.796 $\pm$ 0.012	0.718 $\pm$ 0.013	0.542 $\pm$ 0.020	0.798 $\pm$ 0.020	0.713 $\pm$ 0.015	0.549 $\pm$ 0.013	0.767 $\pm$ 0.006	0.661 $\pm$ 0.011	0.477 $\pm$ 0.013
Focal loss	0.794 $\pm$ 0.010	0.716 $\pm$ 0.010	0.543 $\pm$ 0.017	0.796 $\pm$ 0.020	0.711 $\pm$ 0.018	0.537 $\pm$ 0.016	0.765 $\pm$ 0.008	0.661 $\pm$ 0.014	0.477 $\pm$ 0.008
CFAL	0.802 $\pm$ 0.014	0.723 $\pm$ 0.006	0.542 $\pm$ 0.026	0.801 $\pm$ 0.019	0.727 $\pm$ 0.016	0.561 $\pm$ 0.010	0.769 $\pm$ 0.009	0.666 $\pm$ 0.016	0.477 $\pm$ 0.015
CE + soft F1	0.791 $\pm$ 0.015	0.712 $\pm$ 0.015	0.524 $\pm$ 0.025	0.797 $\pm$ 0.016	0.713 $\pm$ 0.017	0.522 $\pm$ 0.011	0.756 $\pm$ 0.028	0.661 $\pm$ 0.024	0.443 $\pm$ 0.005
CE + soft MCC	0.790 $\pm$ 0.012	0.714 $\pm$ 0.012	0.541 $\pm$ 0.026	0.773 $\pm$ 0.016	0.711 $\pm$ 0.011	0.540 $\pm$ 0.007	0.759 $\pm$ 0.024	0.662 $\pm$ 0.024	0.448 $\pm$ 0.002
Balanced sampling	0.794 $\pm$ 0.014	0.714 $\pm$ 0.013	0.541 $\pm$ 0.028	0.795 $\pm$ 0.016	0.712 $\pm$ 0.018	0.535 $\pm$ 0.022	0.745 $\pm$ 0.017	0.661 $\pm$ 0.016	0.473 $\pm$ 0.011
OKO	0.793 $\pm$ 0.015	0.719 $\pm$ 0.008	0.525 $\pm$ 0.022	0.803 $\pm$ 0.018	0.719 $\pm$ 0.018	0.534 $\pm$ 0.019	0.783 $\pm$ 0.011	0.671 $\pm$ 0.022	0.467 $\pm$ 0.025
ProGAN augmentation	0.792 $\pm$ 0.007	0.728 $\pm$ 0.016	0.543 $\pm$ 0.037	0.800 $\pm$ 0.015	0.712 $\pm$ 0.022	0.576 $\pm$ 0.018	0.770 $\pm$ 0.005	0.658 $\pm$ 0.013	0.459 $\pm$ 0.037

Table 7: Patch-feature tier-resolved test macro F1 (head/body/tail support tiers) per imbalance parameter, mean $\pm$ standard deviation across seeds.

Method	$\lambda = 0.5$			$\lambda = 1.0$			$\lambda = 1.5$		
	Head	Body	Tail	Head	Body	Tail	Head	Body	Tail
Weighted CE	0.870 $\pm$ 0.009	0.799 $\pm$ 0.006	0.706 $\pm$ 0.056	0.870 $\pm$ 0.011	0.797 $\pm$ 0.024	0.686 $\pm$ 0.033	0.845 $\pm$ 0.015	0.739 $\pm$ 0.026	0.602 $\pm$ 0.064
Focal loss	0.873 $\pm$ 0.012	0.794 $\pm$ 0.006	0.706 $\pm$ 0.052	0.876 $\pm$ 0.019	0.798 $\pm$ 0.020	0.698 $\pm$ 0.038	0.833 $\pm$ 0.013	0.722 $\pm$ 0.037	0.555 $\pm$ 0.066
Balanced sampling	0.861 $\pm$ 0.012	0.791 $\pm$ 0.007	0.659 $\pm$ 0.045	0.863 $\pm$ 0.006	0.792 $\pm$ 0.030	0.692 $\pm$ 0.047	0.829 $\pm$ 0.015	0.727 $\pm$ 0.042	0.598 $\pm$ 0.074
RankMix	0.866 $\pm$ 0.019	0.804 $\pm$ 0.012	0.699 $\pm$ 0.046	0.874 $\pm$ 0.018	0.802 $\pm$ 0.018	0.701 $\pm$ 0.031	0.862 $\pm$ 0.008	0.753 $\pm$ 0.025	0.636 $\pm$ 0.036
SC-MIL	0.862 $\pm$ 0.008	0.794 $\pm$ 0.002	0.674 $\pm$ 0.065	0.861 $\pm$ 0.016	0.795 $\pm$ 0.015	0.698 $\pm$ 0.033	0.848 $\pm$ 0.003	0.741 $\pm$ 0.021	0.643 $\pm$ 0.030
MDE-MIL	0.875 $\pm$ 0.014	0.817 $\pm$ 0.008	0.711 $\pm$ 0.056	0.879 $\pm$ 0.005	0.820 $\pm$ 0.011	0.738 $\pm$ 0.050	0.845 $\pm$ 0.009	0.735 $\pm$ 0.022	0.610 $\pm$ 0.024

Table 8: WSI-bag tier-resolved test macro F1 per imbalance parameter, as in Table 7.

edges down to ≈ 0.77 at $\lambda = 1.5$, and WSI-bag head F1 stays near 0.87—whereas tail-tier F1 falls substantially: from ≈ 0.53 to ≈ 0.46 in the patch regime and from ≈ 0.69 to ≈ 0.61 in the WSI-bag regime. The head–tail gap therefore widens with λ , and mitigation methods differ almost entirely in how well they protect the tail rather than the already-saturated head. In the WSI-bag regime at $\lambda = 1.5$, the bag-composition methods retain the highest tail F1 (SC-MIL 0.643, RankMix 0.636), MDE-MIL’s tail erodes to 0.610 despite its strong head and body tiers—mirroring its aggregate erosion at high severity—and focal loss is weakest in the tail (0.555). These tail-tier differences are small relative to their seed-to-seed standard deviations (up to ≈ 0.07), so they should be read as numerical rather than statistically separated.

The three-tier partition (head/body/tail of eight classes each) is designed for the 32-class TCGA-UT label set and does not segment a seven-class label set. All seven BRACS subtypes fall into the head tier; body and tail columns are empty throughout Tables 7 and 8 for the BRACS rows. The tier-resolved analysis is therefore not informative for BRACS, and cross-dataset comparisons of tier F1 are not drawn.

Method	$\lambda = 0.5$		$\lambda = 1.0$		$\lambda = 1.5$		T
	ECE	ECE+TS	ECE	ECE+TS	ECE	ECE+TS	
Weighted CE	0.156 $\pm$ 0.003	0.015 $\pm$ 0.002	0.157 $\pm$ 0.013	0.016 $\pm$ 0.005	0.174 $\pm$ 0.008	0.011 $\pm$ 0.006	2.43 $\pm$ 0.01
Focal loss	0.098 $\pm$ 0.003	0.010 $\pm$ 0.001	0.100 $\pm$ 0.012	0.012 $\pm$ 0.002	0.150 $\pm$ 0.008	0.013 $\pm$ 0.004	1.71 $\pm$ 0.22
CFAL	0.723 $\pm$ 0.009	0.067 $\pm$ 0.007	0.728 $\pm$ 0.021	0.070 $\pm$ 0.008	0.695 $\pm$ 0.013	0.077 $\pm$ 0.006	0.05 $\pm$ 0.00
CE + soft F1	0.179 $\pm$ 0.007	0.016 $\pm$ 0.004	0.184 $\pm$ 0.012	0.014 $\pm$ 0.002	0.217 $\pm$ 0.017	0.013 $\pm$ 0.003	3.22 $\pm$ 0.26
CE + soft MCC	0.162 $\pm$ 0.001	0.012 $\pm$ 0.004	0.169 $\pm$ 0.008	0.013 $\pm$ 0.002	0.183 $\pm$ 0.019	0.015 $\pm$ 0.005	2.51 $\pm$ 0.03
Balanced sampling	0.156 $\pm$ 0.005	0.016 $\pm$ 0.002	0.158 $\pm$ 0.013	0.014 $\pm$ 0.003	0.182 $\pm$ 0.011	0.017 $\pm$ 0.004	2.37 $\pm$ 0.05
OKO	0.071 $\pm$ 0.011	0.030 $\pm$ 0.003	0.193 $\pm$ 0.006	0.041 $\pm$ 0.004	0.192 $\pm$ 0.013	0.037 $\pm$ 0.007	0.70 $\pm$ 0.08
ProGAN augmentation	0.189 $\pm$ 0.001	0.014 $\pm$ 0.004	0.192 $\pm$ 0.017	0.019 $\pm$ 0.002	0.224 $\pm$ 0.009	0.019 $\pm$ 0.006	4.05 $\pm$ 0.08

Table 9: Patch-feature calibration: ECE before (raw) and after (TS) temperature scaling across evaluated imbalance parameters. ECE is reported per λ , whereas the single T column is one fitted temperature averaged across seeds and all three severities and is shown only as a coarse summary; the per- λ temperatures actually applied appear in Table 13, along with full NLL and Brier metrics.

Probability Calibration. In the patch regime, CFAL is a calibration outlier: its affinity-based outputs are nearly flat over 32 classes, requiring strong temperature concentration ($T \approx$

Method	$\lambda = 0.5$		$\lambda = 1.0$		$\lambda = 1.5$		T
	ECE	ECE+TS	ECE	ECE+TS	ECE	ECE+TS	
Weighted CE	0.099 ± 0.012	0.022 ± 0.007	0.092 ± 0.010	0.022 ± 0.014	0.109 ± 0.009	0.023 ± 0.003	1.98 ± 0.05
Focal loss	0.049 ± 0.013	0.019 ± 0.004	0.054 ± 0.008	0.016 ± 0.005	0.105 ± 0.020	0.031 ± 0.004	1.46 ± 0.14
Balanced sampling	0.105 ± 0.010	0.025 ± 0.007	0.099 ± 0.012	0.020 ± 0.004	0.124 ± 0.020	0.027 ± 0.003	2.05 ± 0.06
RankMix	0.025 ± 0.015	0.020 ± 0.009	0.026 ± 0.005	0.018 ± 0.005	0.065 ± 0.008	0.026 ± 0.004	1.16 ± 0.15
SC-MIL	0.096 ± 0.006	0.026 ± 0.005	0.093 ± 0.011	0.018 ± 0.004	0.110 ± 0.001	0.024 ± 0.006	1.90 ± 0.04
MDE-MIL	0.051 ± 0.007	0.037 ± 0.005	0.042 ± 0.006	0.034 ± 0.007	0.044 ± 0.004	0.039 ± 0.010	0.96 ± 0.10

Table 10: WSI-bag calibration: ECE (per λ) and a single fitted temperature T averaged across seeds and severities, shown only as a coarse summary. Columns as in Table 9; per- λ temperatures appear in Table 14.

0.05) and remaining substantially miscalibrated ($\text{ECE} \approx 0.07$) even after scaling—roughly five times higher than all other patch methods post-scaling. All other patch methods achieve ECE below 0.04 after temperature scaling. OKO shows a non-monotone calibration trajectory: well-calibrated at mild severity ($\text{ECE} \approx 0.07$ raw at $\lambda = 0.5$) but deteriorating to $\text{ECE} \approx 0.19$ at moderate severity before scaling. ProGAN augmentation requires the highest temperature correction of any patch method ($T \approx 4.05$), consistent with the highly peaked, overconfident predictions that synthetic-feature training tends to produce; post-scaling ECE is unremarkable ($\text{ECE}+\text{TS} \approx 0.014\text{--}0.019$). In the WSI-bag regime, RankMix is already well calibrated before scaling ($\text{ECE} \approx 0.025$); MDE-MIL shows only marginal improvement from temperature scaling and retains residual miscalibration ($\text{ECE}+\text{TS} \approx 0.037\text{--}0.039$). All WSI-bag methods reach ECE at or below 0.039 after scaling. Calibration rankings are broadly stable across imbalance severities.

BRACS calibration is substantially harder than TCGA-UT. On the patch regime, post-temperature-scaling ECE ranges from 0.041 to 0.110 depending on method and severity, compared to below 0.04 on TCGA-UT; CFAL again shows the lowest temperature ($T \approx 0.07$) and OKO the next lowest ($T \approx 0.20$), consistent with their behaviour on TCGA-UT. On the WSI-bag regime, residual post-scaling ECE ranges from 0.096 to 0.171—three to four times higher than on TCGA-UT—with RankMix the best calibrated before scaling ($\text{ECE} \approx 0.179\text{--}0.235$) and Weighted CE the worst after scaling ($\text{ECE}+\text{TS} \approx 0.127\text{--}0.164$). The elevated post-scaling ECE across all BRACS methods reflects the smaller training pool and the inherent difficulty of the seven-class fine-grained subtyping problem rather than a systematic failure of any one approach.

6 Discussion

The TCGA-UT controlled construction reveals two complementary patterns. First, method rankings are broadly preserved across imbalance severities, with moderate reshuffling among closely performing methods at the extremes. Second, balanced accuracy declines monotonically with λ for almost all methods, with a total range of eight to nine percentage points from the mildest to the strongest constructed regime. These patterns suggest that the relative utility of mitigation methods is largely robust to the particular severity chosen within this range, while absolute performance degrades substantially as rare-class support diminishes.

At the patch level, declines span from approximately 7.8 percentage points for OKO to 10.5 for ProGAN augmentation. ProGAN leads all patch methods at the mildest severity ($\lambda = 0.5$, BAcc 0.729) but drops to last at the harshest ($\lambda = 1.5$, BAcc 0.624)—a reversal that reflects the difficulty of training patch-level generative models when tail-class support falls to single digits: synthetic features complement real training data at mild imbalance but cannot compensate for critically scarce patches at severe imbalance. OKO and CE+soft-F1 maintain their relative advantage at the harshest severity; focal loss is typically among the weakest of the remaining

methods, by margins within seed-to-seed variation.

In the WSI-bag regime, RankMix stands out for its stability, declining by only 6.5 percentage points from $\lambda = 0.5$ to $\lambda = 1.5$. Focal MIL is the least stable, dropping by nearly 12 percentage points and landing at the bottom of the WSI rankings at $\lambda = 1.5$. MDE-MIL declines more than RankMix and falls narrowly below Weighted CE at the harshest regime, showing that consistency regularisation provides diminishing returns as rare-class support becomes critically scarce. RankMix is consistently among the strongest WSI-bag methods and the most robust at $\lambda = 1.5$; SC-MIL stays competitive but does not uniformly beat loss-based baselines—at $\lambda = 0.5$ it trails Weighted CE and focal loss—so the advantage of WSI-specific bag-composition is clear for RankMix but only partial for SC-MIL.

Calibration analysis complements discrimination: CFAL leads on patch discrimination but requires the strongest temperature correction and retains the highest post-scaling ECE. In the WSI-bag regime, RankMix is already well calibrated before scaling, while MDE-MIL shows only marginal improvement from temperature scaling despite leading on discrimination at mild and moderate severities.

The BRACS results refine rather than contradict the TCGA-UT findings. On the patch regime, OKO’s advantage generalises: it leads at all three BRACS severity levels and is the only method that numerically separates from the cluster, while CE+soft-F1 and CFAL are the next most stable—the same ordering as on TCGA-UT. The absolute severity decline on BRACS (2–4 pp) is roughly half that on TCGA-UT (8–11 pp), consistent with the narrower achievable imbalance range (3:1 to 14:1 vs 6:1 to 122:1) and the smaller pool. ProGAN augmentation does not reverse to worst at $\lambda = 1.5$ on BRACS, unlike on TCGA-UT, suggesting that the reversal requires tail-class support to fall below a critical threshold (single-digit patches) that the BRACS pool does not reach.

The WSI-bag regime, however, does not replicate TCGA-UT conclusions. The non-monotone severity responses and high between-seed variance (± 0.09 – 0.10) indicate that the 108-slide training pool is too shallow for the severity signal to be stable across seeds. RankMix’s severity robustness on TCGA-UT does not replicate at this scale. This suggests a minimum pool-depth requirement for the constructed benchmark to produce interpretable rankings: the 1,132-slide TCGA-UT pool supports clean comparisons, while the 108-slide BRACS pool amplifies seed-to-seed noise beyond the severity effect.

Taken together, the TCGA-UT and BRACS results show that patch-regime method rankings generalise to the organ-specific lesion-subtyping setting when the training pool is comparable in scale to the imbalance range imposed. The prediction unit remains the dominant determinant of absolute performance: WSI-bag methods achieve balanced accuracy approximately ten percentage points higher than patch methods on TCGA-UT at equivalent severity. On BRACS, where pool depth limits the severity range achievable, the WSI-bag and patch regimes converge to a narrower absolute performance gap, and the WSI-bag method ordering becomes less stable.

7 Conclusion

This report defines class-imbalance benchmarks for both patch-level and WSI-bag computational pathology. The TCGA-UT component keeps all 32 cancer types, preserves patient-disjoint validation and test splits, and varies only the training-support frequency under native-prevalence class ordering at $\lambda \in \{0.5, 1.0, 1.5\}$. The shared pool of 1,132 slides is determined by the mildest parameter $\lambda = 0.5$, whose flat profile saturates the rarest class’s available supply earliest. The three regimes span head-to-tail ratios of approximately 6:1, 30:1, and 122:1, bracketing and extending the native TCGA-UT ratio ($\approx 28:1$).

The TCGA-UT results show that method rankings are broadly preserved across all three constructed severities, while balanced accuracy declines by eight to eleven percentage points from the mildest to the strongest regime. At the patch level, ProGAN augmentation leads at

mild severity ($\lambda = 0.5$, BAcc 0.729) and undergoes the steepest decline (10.5 pp), reversing from best to worst as tail-class support diminishes; OKO and CE+soft-F1 are most robust at the harshest severity. At the WSI-bag level, RankMix is consistently among the strongest methods and is the most severity-robust; SC-MIL remains competitive but does not uniformly outperform loss-based baselines, and MDE-MIL’s advantage over those baselines erodes under the most extreme imbalance. The prediction unit remains the dominant performance determinant: WSI-bag methods achieve balanced accuracy roughly 12 percentage points above patch-feature methods at equivalent imbalance.

The BRACS component extends the benchmark design to seven-class breast-lesion subtyping with patient-disjoint splits and ROI-derived patch and bag evidence. The prepared native distribution (2.25:1) falls below the minimum threshold for a meaningful native imbalance benchmark, so BRACS is evaluated under the same three power-law severity levels as TCGA-UT using a 108-slide pool (constructed head-to-tail ratios 3:1, 6:1, and 14:1). On the patch regime, OKO retains its lead across all BRACS severities and the severity decline is smaller than on TCGA-UT, consistent with the narrower constructed range. On the WSI-bag regime, non-monotone severity responses and high between-seed variance (± 0.09 – 0.10) indicate that the 108-slide pool is too shallow for the severity signal to dominate; the WSI-bag method ordering is not stable across severities. These results confirm the patch-regime conclusions from TCGA-UT while establishing a pool-depth requirement for the WSI-bag benchmark to produce interpretable severity comparisons.

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A Constructed Training-Support Profiles

λ	Training slides	Support range	Head:tail ratio
0.5	1132	20–112	$\approx 6:1$
1.0	1132	9–272	$\approx 30:1$
1.5	1132	4–487	$\approx 122:1$
native (full pool)	6110	20–567	$\approx 28:1$

Table 11: Constructed training-support profiles under native-prevalence ordering. Support range and head:tail ratio are the minimum and maximum per-class slide counts averaged across seeds. The native full-pool row is shown for reference; it is not part of the constructed experiment.

λ	Training slides	Support range	Head:tail ratio
0.5	108	10–26	$\approx 3:1$
1.0	108	7–40	$\approx 6:1$
1.5	108	4–55	$\approx 14:1$
native (full pool)	636	67–142	$\approx 2:1$

Table 12: BRACS power-law constructed training-support profiles. Pool size 108 WSIs; native full-pool row shown for reference.

B Full Calibration Metrics

Method	λ	NLL	NLL+TS	Brier	Brier+TS	ECE	ECE+TS	T
Balanced sampling	$\lambda = 0.5$	1.314 ± 0.117	0.839 ± 0.039	0.390 ± 0.013	0.339 ± 0.012	0.156 ± 0.005	0.016 ± 0.002	2.441 ± 0.095
Balanced sampling	$\lambda = 1.0$	1.301 ± 0.051	0.841 ± 0.051	0.391 ± 0.033	0.339 ± 0.026	0.158 ± 0.013	0.014 ± 0.003	2.334 ± 0.045
Balanced sampling	$\lambda = 1.5$	1.620 ± 0.112	1.060 ± 0.048	0.457 ± 0.023	0.399 ± 0.013	0.182 ± 0.011	0.017 ± 0.004	2.334 ± 0.045
CE + soft F1	$\lambda = 0.5$	1.626 ± 0.137	0.879 ± 0.044	0.413 ± 0.015	0.343 ± 0.011	0.179 ± 0.007	0.016 ± 0.004	2.884 ± 0.112
CE + soft F1	$\lambda = 1.0$	1.819 ± 0.050	0.889 ± 0.050	0.417 ± 0.032	0.343 ± 0.024	0.184 ± 0.012	0.014 ± 0.002	3.258 ± 0.124
CE + soft F1	$\lambda = 1.5$	2.274 ± 0.170	1.047 ± 0.060	0.486 ± 0.038	0.391 ± 0.029	0.217 ± 0.017	0.013 ± 0.003	3.520 ± 0.067
CE + soft MCC	$\lambda = 0.5$	1.375 ± 0.041	0.846 ± 0.017	0.397 ± 0.009	0.341 ± 0.009	0.162 ± 0.001	0.012 ± 0.004	2.552 ± 0.085
CE + soft MCC	$\lambda = 1.0$	1.392 ± 0.045	0.874 ± 0.039	0.411 ± 0.026	0.353 ± 0.023	0.169 ± 0.008	0.013 ± 0.002	2.496 ± 0.095
CE + soft MCC	$\lambda = 1.5$	1.713 ± 0.148	1.053 ± 0.059	0.451 ± 0.041	0.392 ± 0.030	0.183 ± 0.019	0.015 ± 0.005	2.497 ± 0.128
CFAL	$\lambda = 0.5$	3.232 ± 0.006	0.995 ± 0.042	0.952 ± 0.000	0.352 ± 0.014	0.723 ± 0.009	0.067 ± 0.007	0.050 ± 0.000
CFAL	$\lambda = 1.0$	3.228 ± 0.011	0.997 ± 0.087	0.951 ± 0.001	0.349 ± 0.029	0.728 ± 0.021	0.070 ± 0.008	0.050 ± 0.000
CFAL	$\lambda = 1.5$	3.246 ± 0.002	1.165 ± 0.049	0.953 ± 0.000	0.394 ± 0.018	0.695 ± 0.013	0.077 ± 0.006	0.050 ± 0.000
Focal loss	$\lambda = 0.5$	0.933 ± 0.040	0.819 ± 0.025	0.355 ± 0.010	0.336 ± 0.010	0.098 ± 0.003	0.010 ± 0.001	1.565 ± 0.030
Focal loss	$\lambda = 1.0$	0.933 ± 0.055	0.818 ± 0.053	0.357 ± 0.031	0.337 ± 0.027	0.100 ± 0.012	0.012 ± 0.002	1.548 ± 0.052
Focal loss	$\lambda = 1.5$	1.315 ± 0.076	0.970 ± 0.045	0.417 ± 0.017	0.373 ± 0.015	0.150 ± 0.008	0.013 ± 0.004	2.020 ± 0.000
OKO	$\lambda = 0.5$	0.931 ± 0.029	0.886 ± 0.031	0.352 ± 0.012	0.343 ± 0.012	0.071 ± 0.011	0.030 ± 0.003	0.804 ± 0.016
OKO	$\lambda = 1.0$	1.083 ± 0.076	0.902 ± 0.074	0.383 ± 0.028	0.337 ± 0.029	0.193 ± 0.006	0.041 ± 0.004	0.636 ± 0.012
OKO	$\lambda = 1.5$	1.188 ± 0.040	1.019 ± 0.047	0.410 ± 0.019	0.365 ± 0.022	0.192 ± 0.013	0.037 ± 0.007	0.651 ± 0.000
ProGAN augmentation	$\lambda = 0.5$	2.166 ± 0.069	0.855 ± 0.012	0.414 ± 0.002	0.332 ± 0.002	0.189 ± 0.001	0.014 ± 0.004	3.978 ± 0.077
ProGAN augmentation	$\lambda = 1.0$	2.226 ± 0.230	0.866 ± 0.074	0.419 ± 0.037	0.334 ± 0.027	0.192 ± 0.017	0.019 ± 0.002	4.024 ± 0.157
ProGAN augmentation	$\lambda = 1.5$	2.745 ± 0.185	1.032 ± 0.060	0.485 ± 0.020	0.383 ± 0.019	0.224 ± 0.009	0.019 ± 0.006	4.159 ± 0.080
Weighted CE	$\lambda = 0.5$	1.303 ± 0.074	0.830 ± 0.027	0.387 ± 0.012	0.335 ± 0.011	0.156 ± 0.003	0.015 ± 0.002	2.441 ± 0.095
Weighted CE	$\lambda = 1.0$	1.297 ± 0.078	0.825 ± 0.056	0.387 ± 0.034	0.334 ± 0.028	0.157 ± 0.013	0.016 ± 0.005	2.413 ± 0.047
Weighted CE	$\lambda = 1.5$	1.530 ± 0.086	0.969 ± 0.043	0.431 ± 0.016	0.371 ± 0.015	0.174 ± 0.008	0.011 ± 0.006	2.440 ± 0.047

Table 13: Patch-feature calibration: NLL, Brier score, and ECE before (raw) and after (TS) temperature scaling, with fitted temperature T . One row per method per imbalance parameter.

Method	λ	NLL	NLL+TS	Brier	Brier+TS	ECE	ECE+TS	T
MDE-MIL	$\lambda = 0.5$	0.620 ± 0.021	0.588 ± 0.021	0.226 ± 0.010	0.221 ± 0.010	0.051 ± 0.007	0.037 ± 0.005	0.813 ± 0.032
MDE-MIL	$\lambda = 1.0$	0.582 ± 0.029	0.580 ± 0.029	0.220 ± 0.010	0.220 ± 0.011	0.042 ± 0.006	0.034 ± 0.007	1.049 ± 0.020
MDE-MIL	$\lambda = 1.5$	0.741 ± 0.053	0.740 ± 0.052	0.276 ± 0.013	0.276 ± 0.013	0.044 ± 0.004	0.039 ± 0.010	1.015 ± 0.039
Balanced sampling	$\lambda = 0.5$	0.873 ± 0.054	0.611 ± 0.022	0.271 ± 0.017	0.245 ± 0.012	0.105 ± 0.010	0.025 ± 0.007	1.999 ± 0.101
Balanced sampling	$\lambda = 1.0$	0.850 ± 0.030	0.611 ± 0.030	0.268 ± 0.016	0.246 ± 0.012	0.099 ± 0.012	0.020 ± 0.004	2.021 ± 0.067
Balanced sampling	$\lambda = 1.5$	1.140 ± 0.249	0.790 ± 0.137	0.330 ± 0.050	0.300 ± 0.044	0.124 ± 0.020	0.027 ± 0.003	2.143 ± 0.223
Focal loss	$\lambda = 0.5$	0.602 ± 0.036	0.563 ± 0.026	0.233 ± 0.014	0.228 ± 0.012	0.049 ± 0.013	0.019 ± 0.004	1.310 ± 0.000
Focal loss	$\lambda = 1.0$	0.624 ± 0.062	0.566 ± 0.055	0.233 ± 0.027	0.228 ± 0.025	0.054 ± 0.008	0.016 ± 0.005	1.416 ± 0.027
Focal loss	$\lambda = 1.5$	0.902 ± 0.090	0.744 ± 0.057	0.309 ± 0.034	0.290 ± 0.026	0.105 ± 0.020	0.031 ± 0.004	1.654 ± 0.000
Weighted CE	$\lambda = 0.5$	0.848 ± 0.055	0.600 ± 0.024	0.263 ± 0.018	0.238 ± 0.012	0.099 ± 0.012	0.022 ± 0.007	2.022 ± 0.115
Weighted CE	$\lambda = 1.0$	0.807 ± 0.068	0.582 ± 0.048	0.251 ± 0.025	0.231 ± 0.022	0.092 ± 0.010	0.022 ± 0.014	2.020 ± 0.000
Weighted CE	$\lambda = 1.5$	0.940 ± 0.070	0.708 ± 0.046	0.302 ± 0.024	0.275 ± 0.020	0.109 ± 0.009	0.023 ± 0.003	1.912 ± 0.073
RankMix	$\lambda = 0.5$	0.624 ± 0.045	0.623 ± 0.044	0.240 ± 0.019	0.239 ± 0.018	0.025 ± 0.015	0.020 ± 0.009	1.015 ± 0.052
RankMix	$\lambda = 1.0$	0.573 ± 0.055	0.569 ± 0.051	0.230 ± 0.026	0.229 ± 0.026	0.026 ± 0.005	0.018 ± 0.005	1.097 ± 0.021
RankMix	$\lambda = 1.5$	0.706 ± 0.063	0.654 ± 0.045	0.264 ± 0.016	0.257 ± 0.015	0.065 ± 0.008	0.026 ± 0.004	1.370 ± 0.026
SC-MIL	$\lambda = 0.5$	0.773 ± 0.044	0.577 ± 0.017	0.262 ± 0.010	0.239 ± 0.004	0.096 ± 0.006	0.026 ± 0.005	1.913 ± 0.098
SC-MIL	$\lambda = 1.0$	0.765 ± 0.087	0.586 ± 0.059	0.259 ± 0.031	0.239 ± 0.025	0.093 ± 0.011	0.018 ± 0.004	1.848 ± 0.036
SC-MIL	$\lambda = 1.5$	0.934 ± 0.070	0.693 ± 0.055	0.300 ± 0.012	0.274 ± 0.015	0.110 ± 0.001	0.024 ± 0.006	1.932 ± 0.037

Table 14: WSI-bag calibration: NLL, Brier score, and ECE before and after temperature scaling. Columns as in Table 13.